

USER INFORMATION
READ CAREFULLY

INFUSOL® S3
SODIUM CHLORIDE 3% w/v INTRAVENOUS INFUSION BP

Composition:

Each 100 mL solution contains:

Sodium Chloride BP 3 g

Electrolytes:	mmol/L	mEq/L
Sodium Ion (Na ⁺)	513.3	513.3
Chloride Ion (Cl ⁻)	513.3	513.3
Osmolality :	1026 mOsm/L	

Product Description:

A clear and colourless solution.

Pharmacodynamic:

Mechanism of Action

INFUSOL® S3 is a hypertonic solution of the most abundant ions in the body fluids, namely Sodium and Chloride. Sodium is the primary cation of the extracellular space and together with various anions, regulates the size of this. Sodium and Potassium are the major mediators of bioelectric processes within the body.

The Sodium content and the fluid metabolism of the body are closely coupled to each other. Each deviation of the plasma sodium concentration from the physiological one simultaneously affects the fluid status of the body.

An increase in the Sodium content of the body results also means reduction of the body's free water content independent of the serum osmolality.

Pharmacodynamic Effect

INFUSOL® S3 is mainly used for rapid Sodium and Chloride substitution in situations of serious depletion of those electrolytes.

Pharmacokinetics:

Absorption

As the solutions are administered directly into the circulation, their bioavailability is 100%.

Further Pharmacokinetic Properties

The total Sodium content of the body is ca. 80 mmol/kg of which ca. 97% is extracellular and ca. 3% intracellular. The daily turnover is ca. 100 - 180 mmol (corresponding to 1.5 - 2.5 mmol/kg body weight).

The kidneys are the major regulator of the Sodium and water balances. In co-operation with the hormonal control mechanisms (renin-angiotensin-aldosterone system, antidiuretic hormone) and the hypothetical natriuretic hormone they are primarily responsible for keeping the volume of the extracellular space constant and regulating its fluid composition.

Chloride is exchanged for hydrogen carbonate in the tubule system and is, thus, involved in the regulation of the acid-base balance.

Indication:

Treatment of:

- Hyponatraemia
- Hypochloraemia
- Hypotonic hyperhydration

Recommended Dosage:

The dosage is adjusted according to the actual electrolyte requirements.

A Sodium deficit is calculated according to the following formula: $\text{Na}^+_{\text{required}} \text{ (mmol)} = (\text{Na}^+_{\text{normal}} - \text{Na}^+_{\text{actual}}) \times \text{kg BW} \times 0.2$

(The term "kg BW × 0.2" represents the extracellular volume, BW = body weight)

Adults

A daily Sodium intake of 3 to 6 mmol/kg BW should not be exceeded if possible.

Paediatric population

A daily Sodium intake of 3 to 5 mmol/kg BW should not be exceeded if possible.

Elderly

As for adults. However, attention should be paid to the patient's cardiocirculatory status.

Maximum infusion rate

The maximum infusion rate depends on the clinical situation. In patients with chronic hyponatraemia the infusion rate should be sufficiently low so that the serum Sodium level does not rise faster than $0.35 \text{ mmol} \times \text{l}^{-1} \times \text{h}^{-1}$.

Route of administration: Intravenous (IV)

Contraindications:

- Hyponatraemia
- Hyperchloraemia
- Isotonic hyperhydration
- Hypertonic hyperhydration

Warnings and Precautions:

For central venous infusion only.

The solutions should only be administered with caution to patients with:

- Hypokalaemia
- Diseases requiring restriction of Sodium intake, such as:
 - cardiac insufficiency,
 - generalized oedema
 - pulmonary oedema,
 - hypertension,
 - eclampsia,
 - severe disturbances of kidney function.

In patients with chronic hyponatraemia, a too rapid increase of the serum Sodium level may lead to organic brain damage, the "osmotic demyelination syndrome". Care must be taken to keep the increase of the serum Sodium level within the limit.

It is necessary to monitor the serum electrolyte levels, acid-base and water balance.

For both bottle and bag packaging:

Single dose container. Discard all unused contents. Do not use if leakage is detected. If any visible solid particles, growth or turbidity appears during storage, the product should not be used. Do not reconnect partially used containers. Administer immediately following the insertion of the infusion set.

As with all parenteral solutions, before adding medications, compatibility of these additives with the solution in the container must be assessed. Before introducing an additive verify it is soluble and stable in water. Incompatibility of an additive medication with the Sodium Chloride Solution is to be judged by checking for colour change and/or precipitate, insoluble complexes or crystals that may appear.

Thorough and careful aseptic mixing of any additive is mandatory. Solutions containing additives should be used immediately after preparation, unless preparation has taken place in controlled and validated aseptic conditions. Those additives known to be incompatible should not be used.

When a compatible medication is added to this formulation, the solution must be administered immediately, unless dilution has taken place in controlled and validated aseptic conditions.

Interactions with Other Medicaments:

Therapy with corticosteroids or ACTH may be associated with an increased retention of Sodium and water, possibly leading to oedema and hypertension.

Pregnancy and Lactation:

There are no data from the use of higher concentrated Sodium Chloride solutions for infusion in pregnant and lactating woman. In the intended indications no risks have to be expected, when volume, electrolyte and acid/base levels are carefully monitored. Caution should be exercised in the presence of toxæmia of pregnancy.

Side Effects:

Summary of the Safety Profile

Undesirable effects are mainly caused by inadequate dosages and/or infusion rates. Under conditions of proper administration, they will occur only rarely, i.e. in less than 1/1000 of treated patients.

Listing of Adverse Reactions

Metabolism and Nutrition Disorders

Sodium Chloride administration may lead to an increase of blood Sodium and Chloride levels (hypernatraemia, hyperchloraemia). Rapid infusion may lead to acute volume overload.

Gastrointestinal Disorders

Diarrhoea due to inadequately high infusion rates.

Renal and Urinary Disorders

Osmotically induced polyuria (osmotic diuresis) due to inadequately high infusion rates.

Symptoms and Treatment of Overdose:

Symptoms

Overdose can lead to hyperhydration, hypernatraemia, hyperchloraemia, hyperosmolarity and disturbances of the acid-base balance, in particular acidosis.

Clinical symptoms of hypernatraemia:

Thirst, motor unrest, hyperexcitability of muscles, hypertension, vertigo, vomiting, somnolence, dryness of skin and mucosae, fever, oedema, congestive heart failure, hyperosmolar coma.

Treatment

Immediate stop of infusion, administration of diuretics with continuous monitoring of the electrolyte balance, correction of electrolyte disturbances, counteraction of acidosis.

Shelf Life:

3 years from manufacturing date in proposed storage condition. Do not use after expiry.

Storage condition:

Do not store above 30°C.

Dosage form and packaging available:

500 mL x 20 LDPE plastic bottle per carton

500 mL x 20 PVC Medical plastic collapsible bag per carton

Manufacturer / Product Registration Holder:



AIN MEDICARE SDN.BHD.

Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA.

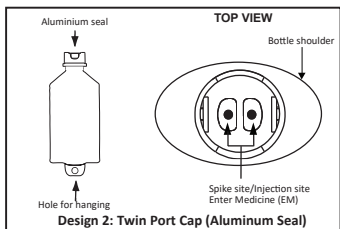
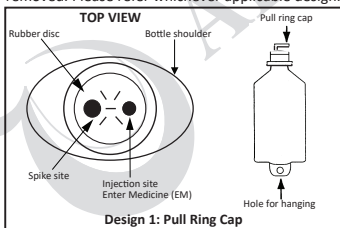
HANDLING INSTRUCTION

AIN MEDICARE INTRAVENOUS INFUSION FLUID IN LOW DENSITY POLYETHYLENE (LDPE) BOTTLE

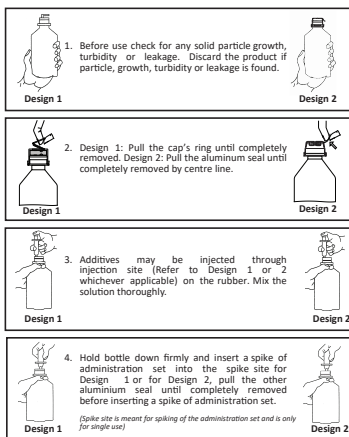
The bottle is made from injectable grade polyethylene.

PARTS OF THE PLASTIC BOTTLE

Top view after pull ring cap and aluminum seal is removed. Please refer whichever applicable design.



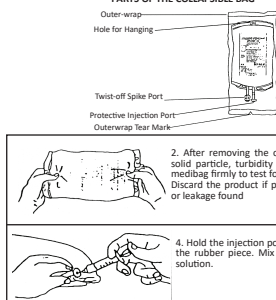
INSTRUCTION FOR THE USAGE OF IV ADMINISTRATION SET & ADDITION OF DRUG



AIN MEDICARE INTRAVENOUS INFUSION FLUID IN PVC MEDICAL COLLAPSIBLE BAG

The intravenous solution in medibag is packed and sterilized in an outerwrap.

PARTS OF THE COLLAPSIBLE BAG



INSTRUCTION FOR THE USAGE OF IV ADMINISTRATION SET & ADDITION OF DRUG

